

An AI-ML model for lake water quality prediction of Indian cities

1. Project description and objectives

Build an AI-ML model for lake water quality prediction of Indian cities: The objective of this project is to quantitatively analyze the relationship between Landsat-8/9 OLI/TIRS spectral reflectance maps (independent variable(s)) and lake/tank water quality parameters, based on the ground truth dataset available for Bangalore lakes/tanks. The proposed workflow includes:

- (a) integrating multi-year multispectral images of the lake/tanks, and ground truth water quality parameters of the lakes/tanks of Bengaluru between July 2023–November 2025 to construct a harmonized, quality-controlled database for robust temporal and spatial analysis;
- (b) developing AI/ML model(s) that combine multi-spectral reflectance data, and water-quality data to predict future pollution levels;
- (d) training, validating, and testing the model(s) using appropriate data-partitioning and cross-validation strategies, with performance evaluated on independent test datasets using standard statistical metrics;
- (e) generating short-term and long-term water-quality forecasts using the trained models; and
- (f) assessing model robustness through spatial evaluation across monitoring locations within cities and testing generalization across multiple cities.

The outlined workflow is indicative, and candidates are encouraged to adopt alternative or improved methodologies as appropriate. Use your imagination!

2. Relevant literature/Starting work

Refer to the attached publications for getting started. The expectations are to use these works as a starting point and build models which can go beyond what is already published.

3. Team formation and communication

Class is divided into groups, and each group will work on Bangalore dataset for analysis. The group consists of up to 5 members and a project communicator. The communicator is responsible for communicating, i.e., uploading the final project report and final presentation.

4. Data sources

Satellite imagery (multispectral):

Please download the required image datasets from **USGS EarthExplorer** for the study period (July 2023–Nov 2025). Use EarthExplorer to search and download suitable multispectral data (Landsat 8/9 OLI/TIRS) and any available high spectral resolution imagery for the lakes/tanks of interest. Perform a spatial search using coordinates of selected lakes and filter by date range to retrieve as many clear observations as possible over the period. Primary source — use EarthExplorer first. Link: <https://earthexplorer.usgs.gov/>

Ground truth water quality data:

Monthly lake water quality reports for Bengaluru lakes for the period July 2023 through November 2025 are available and should be downloaded in full. These reports are published by the Karnataka State Pollution Control Board and include multiple water quality parameters. Link: <https://data.opencity.in/dataset/bengaluru-lake-monthly-water-quality-reports>

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5. Tools & environment

1. **Google Colab** (free, GPU not needed + easy sharing).
2. Python stack: **pandas**, **numpy**, **matplotlib/plotly**, **seaborn** (optional), **scikit-learn**, **statsmodels**.
3. Version control: share notebooks via GitHub or Google Drive.

6. Workflow (suggested notebooks)

1. **Data download & preprocessing**
 - o Download multispectral satellite images for selected lakes from USGS EarthExplorer for the study period (July 2023–May 2025).
 - o Perform preprocessing.
 - o Extract lake-level spectral features (band reflectance, band ratios, spectral indices).
 - o Download monthly water quality measurements for the corresponding lakes and time period.
 - o Temporally align satellite data with ground-truth measurements; handle missing or noisy observations.
2. **Exploratory Data Analysis (EDA)**
 - o Time-series analysis of water quality parameters and corresponding spectral features.
 - o Seasonal and inter-annual variability in water quality across lakes.
 - o Band-wise and index-wise sensitivity analysis using scatter plots and boxplots.
 - o Cross-lake comparison of spectral signatures and water quality distributions.
3. **Statistical, ML & Deep Learning modeling**
 - o Correlation analysis (Pearson/Spearman) between spectral features and water quality parameters.
 - o Baseline regression models for water quality retrieval from reflectance.
 - o Deep learning models (e.g., ANN / CNN-based models) trained on multispectral inputs to predict water quality parameters.
 - o Train–validation–test split across time and/or lakes to assess generalization.
4. **Prediction, interpretation & sensitivity analysis**
 - o Predict water quality parameters from unseen multispectral images.
 - o Evaluate prediction performance using RMSE, MAE, R^2 against ground truth.
 - o Identify important spectral bands and indices influencing predictions.
 - o Discuss limitations related to sensor resolution, atmospheric effects, and data availability.

7. Visualizations & presentation

1. Required visuals: reflectance plots, seasonal boxplots, correlation matrix, and model diagnostics.
2. Presentation: **7 minutes of pre-recorded video presentation + 3 minutes live Q&A**. Each slide deck: 12 slides maximum.

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8. Deliverables (per group)

1. Cleaned merged dataset (CSV / notebook): Spectral reflectance and water-quality
2. EDA visuals: PNG/embedded in notebook.
3. Analytical results: correlation tables, regression outputs, feature importances.
4. Short report (2–4 pages): Methods, key findings, limitations.
5. Slide deck: 12 slides maximum
6. Presentation: 7-minute pre-recorded audio/video presentation preferably < 200 MB

9. Outline of grading scheme (out of 100)

1. Data quality & preprocessing: 20
2. EDA & visual clarity: 20
3. Analysis rigor (models/statistics): 25
4. Interpretation & discussion: 20
5. Presentation & teamwork: 15



Machine learning approach for water quality predictions based on multispectral satellite imageries

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ABSTRACT

Water quality analysis is a vital component of the water resources management and has to be undertaken promptly to make sure environmental regulations are being followed and to eliminate any pollution that could harm the ecosystem. The main objective of this study to retrieve and map the water quality parameters from Sentinel-2 and ResourceSat-2 [Linear Imaging Self-Scanning Sensor (LISS)-IV] multi-spectral satellite data, using Support Vector Machines (SVM), Random Forests (RF), and Multi-Linear regression (MLR) models. This study represents the first attempt to demonstrate the applicability and performance of high-spatial resolution ResourceSat-2 remote sensing satellite's LISS-4 sensor, which operates in three spectral bands in the Visible and Near Infrared Region (VNIR), to predict water quality. Spectral bands of each satellite were used as independent parameter to generate the algorithms for pH, Dissolved Oxygen (DO), Total Suspended Solids (TSS) and Total Dissolved Solids (TDS). The model performance was evaluated based on coefficient of determination (R^2), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Root Mean Square Error (RMSE) statistical indices. The result of this study indicates that the SVM yielded the highest accuracy followed by the RF and MLR. The R^2 , MAE, MAPE and RMSE ranged between 0.78 and 0.99, 0.049–0.24, 0.01–10.9 % and 0.05–0.28 respectively for all the four SVM models across both the sensors. Based on the spatial trend Sentinel-2 was found to be slightly superior to the ResourceSat-2 (LISS-IV) for the estimation of water quality parameters owing to its superior spectral and radiometric resolution, nevertheless ResourceSat-2 (LISS-IV) has its own advantage in terms of high spatial resolution. The results of this study highlight the high potential of machine learning models in conjunction with multispectral satellite images to manage water quality.

1. Introduction

An essential resource for both the environment and people is water (Tzanakakis et al., 2020). Around the world, freshwaters make up around 3 % of the total water resource, but only 5 % of them are usable by humans (Usharani et al., 2010). It is estimated that 40 % of people on Earth lack access to clean water; this number is thought to be rising due to population expansion. The quality of surface water is under threat due to human-induced environmental changes, making it impossible to meet the Sustainable Development Goals set forth by the UN regarding water quality (Campos et al., 2022). In terms of impact, the water crisis-which includes water scarcity, pollution, and other problems is ranked fifth out of the top ten hazards in the Global hazards Report 2020. Water

pollution, in specific, has evolved to be a challenging and complicated issue that threatens the technological and economic strategies of many nations (Tang et al., 2022). River and lake water ecosystem services support aquatic ecosystems and human well-being either directly or indirectly (Ferreira et al., 2023). The global situation is greatly concerned about the growing contamination of water sources, since the majority of water bodies worldwide serve as supplies of water for household and human use (Lin et al., 2022). The water quality metric, which takes into account the physical, chemical, and biological aspects, determines the wellbeing of the aquatic ecosystem which includes lakes and rivers (Beggel et al., 2022). Lakes distinct from rivers in several ways, including the fact that they act as nutrient sinks, have slower flow rates, and longer retention periods (Janicka et al., 2022). Because of its

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extended retention duration, nutrient enrichment in lakes can be lethal from both natural and man-made causes. Numerous factors influence the amount of nutrients in lakes. Changes in the levels of dissolved oxygen, nitrogen, phosphorus, and other chemicals can lead to eutrophication (Akinnawo, 2023). The increased biomass of phytoplankton, the growth of benthic and epiphytic algae, the loss of fisheries, the reduction of habitat diversity, the increase in odour problems caused by an increase in mineral contents, and the depletion of dissolved oxygen in lakes are all among the high potential effects of eutrophication (Costa et al., 2018). As a result of this, one specific issue with water quality monitoring is a complex one that requires analysing a lot of associated measures of variables (Gholizadeh et al., 2016). An increase in anthropogenic activities, including natural influences, is the cause of the high variability among the variables (Mwanake et al., 2023). The thorough assessment of water quality at the regional and national levels is heavily reliant on data from water pollution monitoring (Swain and Sahoo, 2017). Assessing the water environment efficiently and precisely is therefore vital. *In-situ* measurements have historically been able to provide exact water quality parameters by employing specific sampling points; but, large-scale monitoring is limited by the time, labour, and high costs associated with this method (Schaeffer et al., 2013). Furthermore, sampling techniques and laboratory processes may have an impact on the quality of field-based data (Lloyd et al., 2022). The spatial distribution and dynamic variations in water quality components have been reflected by remote sensing technology because of its advantages in terms of both temporal and spatial coverage (Malahlela, 2019; Shi et al., 2018). A variety of spaceborne sensors with visible, infrared, and microwave wavelengths can be utilised to monitor water quality because of high frequency data collecting and large-scale coverage. Since remote sensing data rely on the weather, empirical algorithms based on remote sensing must be precisely calibrated with respect to the research region (Jepsen et al., 2021). As, remote sensing can give very cost-effective spatial and multitemporal information about the uppermost layers of water bodies, it has been employed to monitor water quality parameters in various lakes and reservoirs. Integrative methods based on conventional field-based data collection and remote sensing are viewed as appropriate for water resource surveillance programs since remote sensing data are publicly available (Muhoyi et al., 2022). The benefits of extended time series, large-scale simultaneous observation, and historical backtracking in satellite remote sensing technology can compensate for the drawbacks of *in-situ* surveys and allow the detection of spatiotemporal fluctuations in water quality.

In order to enable swift detection of changes and trends in water quality indicators, modelling of water quality based on process-based hydrodynamic-water quality models have been widely applied (Mardani et al., 2020; Mohammed et al., 2019). However, recent developments in remote sensing sensors and prediction using artificial intelligence, deep learning and machine learning based algorithms are currently popular methods (Bui et al., 2020; Chen et al., 2020; Kong et al., 2019; Tang et al., 2023; Yin et al., 2021; Yu et al., 2024). For simulating and forecasting temperature dynamics and constituent movement in surface water bodies such lakes, rivers, and estuaries, process-based hydrodynamic and water quality models have been used extensively (Mohammed et al., 2021). Process-based hydrodynamic and water quality simulations are hard to calibrate and frequently need high levels of expertise to implement due to their complex model structures and high computational requirements (Baracchini et al., 2020; Chen et al., 2016). This limits their use in water resource management. Furthermore, there are not many common uses for process-based models in near-real-time simulations in water source reservoirs. The applicability of process-based hydrodynamics and water quality models in comprehending the physical mechanisms and variables affecting the thermal dynamics and water quality properties of surface waterbodies is undeniable. Empirical analysis and bio-optical analysis are two distinct types of techniques for assessing water quality indicators using data from satellite remote sensing (Dong et al., 2023; Sagan et al., 2020).

Aquatic remote sensing requires consideration of atmospheric correction, and while bio-optical modelling has made significant advancements in recent years, it is severely limited by data restrictions and challenges with atmospheric correction (Li et al., 2021). A portion of the scientific community has been exploring direct modelling techniques with satellite reflectance data. It is widely recognised that there can be a substantial correlation between remote sensing techniques and optically active parameters (Zhao et al., 2022; Zhu et al., 2022). Yet, due of their intricate interaction between the different components in water, non-optically active parameters can also be quantitatively recovered by remote sensing (Lan et al., 2023; Zhao et al., 2022). Machine learning algorithms have been used more often in the assessment of water quality in recent years due to the advancements in the field of artificial intelligence (Peterson et al., 2020). In order to provide a generic and optimal method for water quality parameter identification, machine learning models can reveal underlying complex nonlinear interactions (Shams et al., 2024). There is a steady increase in its use in modelling and detecting water quality (Arias-Rodriguez et al., 2020). Support Vector Machine Regression (SVR) and Random Forest Regression (RF) are two popular machine learning techniques for evaluating the quality of water (Peterson et al., 2020).

Random Forest (RF), Artificial Neural Network (ANN), Multiple Linear Regression (MLR), Ridge Regression (RR), Adaptive Boosting (AdaBoost), Support Vector Regression (SVR), and eXtreme Gradient Boosting tree (XGBoost) are the machine learning models that are most frequently used for optical water quality parameters. Adusei et al. (2021) forecasted the pH, turbidity, dissolved oxygen (DO), alkalinity, and total dissolved solids (TDS) in the Owabi Dam reservoir using MLR, RF, SVM, and Sentinel satellite imagery. The research found that RF performed better than MLR and SVM among the three algorithms utilised in this study. Kim et al. (2014) used random forest and support vector regression (SVR) to estimate the concentrations of suspended particulate matter (SPM) and chlorophyll-a (Chl-a), two optical water quality indicators, in coastal environments. They found that SVR performed better than the other two machine learning techniques. Muhoyi et al. (2022) developed multitemporal spatialized maps of the concentrations of total suspended solids (TSS), total nitrogen (TN), total phosphorus (TP), and chemical oxygen demand (COD) using empirical equations and satellite data obtained from Sentinel-2. Zheng et al. (2024) used the XGBoost machine learning algorithm to demonstrate an enhanced water quality index for the assessment of water quality Yopurga landfill, China. Using a machine learning approach, gradient boosting outperformed the random forest and support vector machines for the prediction of water quality in Mirpurkhas, Sind (Abbas et al., 2024).

Among the most important water resources on Earth's surface are lakes and wetlands, which are regarded as important ecohydrological units. Loktak Lake is home to a wide range of wetland plants and animals, some of which are internationally threatened, and is rich in biodiversity. The lake is well-known for its phumdis, or floating islands, which are covered in flora and vary in size and thickness. According to Anand et al. (2021) they are essentially heterogeneous masses of soil, plant, and organic matter at different stages of decomposition. The lake's natural processes depend heavily on the phumdis. As a biological sink of essential nutrients, phumdis play a crucial role in preventing eutrophication and algae blooms (LDA and WISA, 2002). Phumdis therefore regulate the lake's nutrient dynamics and water quality (LDA and WISA, 2002). Research has identified significant pollution problems in the Loktak Lake due to high nitrite and nitrate concentrations, with water quality index ranging from 'very poor' to 'poor' (Laishram et al., 2022; Talukdar et al., 2024). There has been reduction in the area of herbaceous wetlands around the Loktak Lake and it is expected to decline in the near future due to anthropogenic factors (Anand et al., 2021). Damming effect caused due to the barrage in the South-Eastern periphery of the lake has caused ecohydrological deficit in the Loktak Lake sub-catchment (Mahato et al., 2023). Over time, a number of

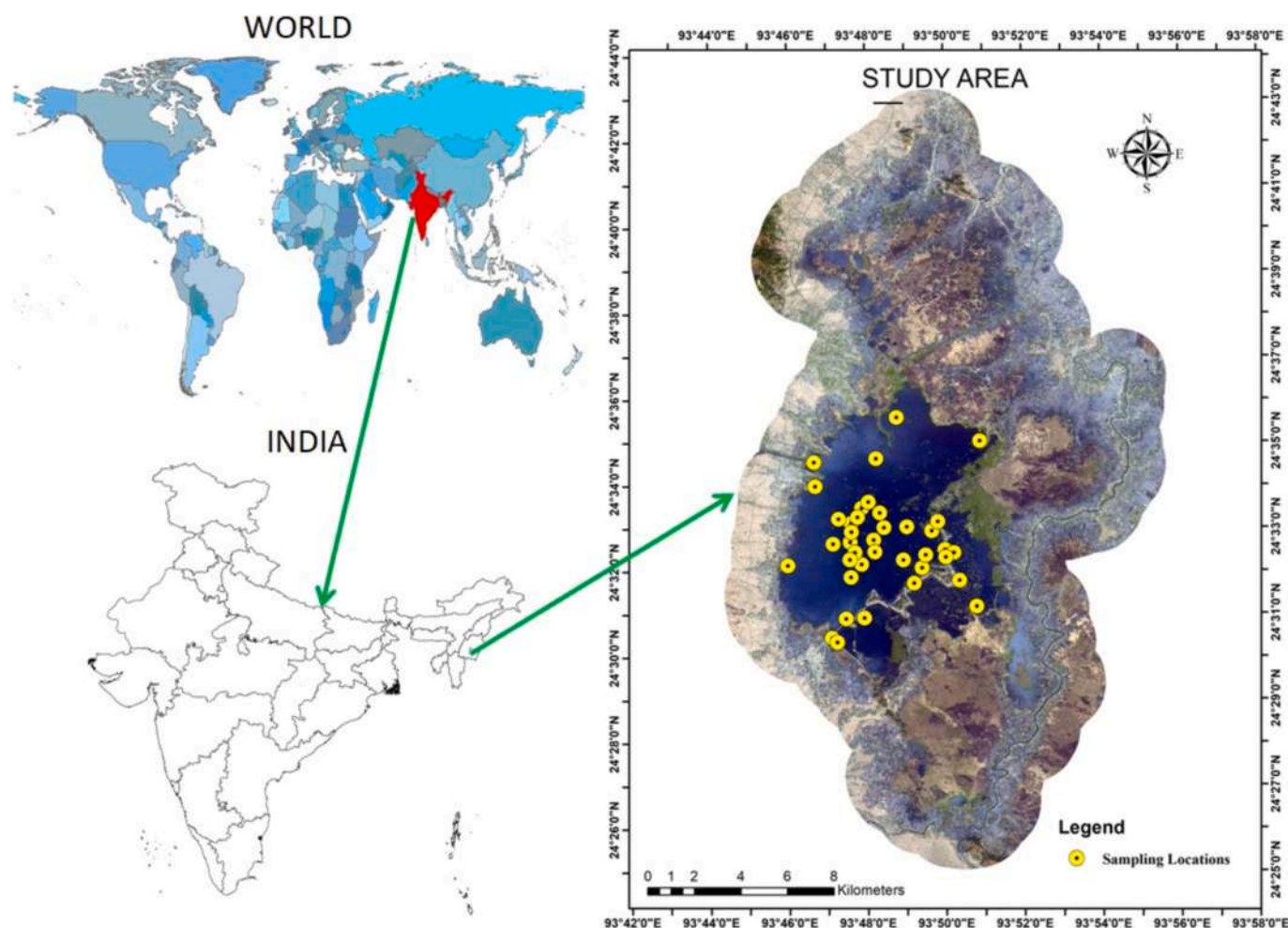


Fig. 1. Study area and sampling locations.

studies have evaluated the water quality of Loktak Lake and have reported on a range of indicators related to water quality (Kangabam et al., 2017; Laishram et al., 2022; Mayanglambam and Neelam, 2020; Roy and Majumder, 2019). Large lakes like Loktak Lake make it impractical and expensive to conduct continuous field monitoring of multiple water quality parameters. These studies have been restricted to *in-situ* measurements to give water quality metrics by using selected sampling points. In the water bodies of North-Eastern India's Inner Himalayan Ranges, there has not been any research done on the prediction of water quality using multispectral satellite imagery and machine learning. Loktak Lake being a Ramsar site in the Inner Himalayan range is a partially or ungauged in terms of both water quality and quantity. The size and the ecological dynamics of such large lake like Loktak Lake possess serious challenges in continuous spatial-temporal gauging. Machine learning algorithms when coupled with satellite remote sensing can uncover underlying complicated connections to provide an encompassing and efficient approach for evaluating water quality parameters for a large area especially in the data scarce regions. The main objective of this research is to provide a dynamic method for predicting the quality of water by employing machine learning algorithms and multi-spectral ResourceSat-2 and Sentinel-2 satellite data. This research also emphasises on the advantages and limitations of resolutions of satellites while comparing the Sentinel-2 with better spectral and radiometric resolution against ResourceSat-2 (LISS IV) with better spatial resolution. The current study is the first attempt to introduce the applicability of deploying ResourceSat-2 remote sensing satellite's high-resolution Linear Imaging Self Scanner (LISS-4) camera operating in

three spectral bands in the Visible and Near Infrared Region (VNIR) novel for prediction of water quality. The research focuses on dynamic monitoring of water quality parameters through high resolution satellite sensors over a large scale. This study was implemented in Loktak Lake located in the North-Eastern part of India. This research offers an excellent scientific basis for enhancing lake pollution control, promoting water pollution reduction, and setting in effective operational methods for monitoring water quality.

2. Materials and methodology

2.1. Study area

Loktak Lake, a freshwater wetland system is situated in Northeastern India's Inner Himalayan Ranges. The Loktak Lake lies between $24^{\circ} 42' N$ - $93^{\circ} 55' E$ and $24^{\circ} 25' N$ - $93^{\circ} 46' E$. The international significance of the lake has led to its designation under the Ramsar Convention (NWA, 2009). According to the Montreux Record, Loktak Lake is one of the 48 wetland areas in the world. Montreux record lists the wetland sites where changes in ecological character have occurred, are occurring, or are likely to occur as a result of technological developments, pollution or other human interference. Although the primary body of water in the lake is 287 km^2 , the catchment as a whole is 5020 km^2 (Anand et al., 2020). The basin experiences 1350 mm of rain on average. The relative humidity varies from 51 to 81% and the annual temperature ranges from $12^{\circ} C$ to $31^{\circ} C$ (Directorate of Environment Government of Manipur, 2013). Nine sub-catchments constitute the Loktak Lake catchment

Table 1
Metadata related to multispectral satellite imageries.

Satellite/Mission	Spatial Resolution	Radiometric Resolution	Bands used	Central Wavelength	Acquisition date
Sentinel-2	10 m	12 bits	B2 (Blue)	493 nm	09 March 2022
			B3 (Green)	560 nm	08 April 2022
			B4 (Red)	665 nm	14 November 2022
			B8 (NIR)	833 nm	02 February 2023
ResourceSat-2 (LISS 4)	5.8 m	10 bits	B2 (Green)	555 nm	09 March 2022
			B3 (Red)	650 nm	04 November 2022
			B4 (NIR)	815 nm	01 February 2023

hydrologically: Khuga, Western, Nambul, Imphal, Kongba, Iril, Thoubal, Heirolk, and Sekmai. The two sub-catchments that feed the lake directly are the Nambul and the Western sub-catchments. Over time, untreated sewage, fertilizers, and plastic pollutants are introduced into the lake by the feeding rivers that flow past large towns and continuously growing farms. The effects of pollution on fish health and mortality are posing a growing danger to the ecosystem of Loktak Lake. Removing the ecological stressors might be the most effective method to restore the wetland and prevent further loss and degradation (Thongam and Meitei, 2021). The Nambul River, which is the primary tributary, has been identified as one of the major environmental stressors that contribute to the wetland's contamination. Therefore, it is essential to regularly monitor the lake and comprehend its three dimensions. The Loktak Lake along with the sampling locations is shown in (Fig. 1).

2.2. Field sampling and satellite data

The field visits were carried out for water quality sampling using portable multi-parameter water quality probe. The field visits were carried out on the days with clear sky approximately synchronous to the pass of the satellites. It was reported that the time gap between the satellite image and the *in-situ* measurements affects the reflectance comparison, it is still reasonable when environmental and water conditions do not undergo rapid changes (Martins et al., 2017). The sampling locations were randomly chosen to cover the maximum extent of the lake covering both the centre and the shores of the lake. Total of 38 sites were monitored between the years 2022–2023. Due to the restrictions imposed by the environment and forest department certain parts of the lake is inaccessible to maintain the biodiversity. Since cloud free days and large storm events would have a substantial impact on the results, field work planning was done to synchronize the field data collection dates with satellite fly over. Surveys were conducted either on date of pass of satellites or with a time frame of one week of satellite pass. Factor of minimal cloud cover over the lake was also considered during the selection of the date of survey. The Sentinel satellite datasets used in this study was downloaded from official portal (Copernicus open access hub) of European Space Agency. The ResourceSat-2 (LISS IV) data were purchased from the Indian Space research Organization (ISRO). The field surveys were carried on 9 March 2022, 8 April 2022, 14 November 2022 and 31 January 2023. The information related image acquisition day of the satellites, resolutions and bands is tabulated in Table 1. The preliminary pre-processing of the Sentinel was done in ERDAS Imagine environment, whereas for LISS-IV was done in Natural Color Image Generator for IRS Satellite Datasets environment. The field surveys were carried out using Hanna HI98194 portable multi-parameter measuring device for the instantaneous onsite measurements of dissolved oxygen (DO), total dissolved solids (TDS) and pH. For the measurement of total suspended solids, the water samples were collected in clean 500 ml bottles. The total suspended solids were measured using Gravimetric analysis in the laboratory. The water sampling was carried out by following the guidelines of Central Pollution Control Board (CPCB), Govt. of India.

2.3. Normalised difference water index (NDWI)

It ought to be emphasised that this lake's extent varies over time and that it reaches its maximum size after the rainy season. Phumdis, also known as floating islands, are heterogeneous masses of soil, vegetation, and organic materials in varying stages of decomposition that float on the water's surface and whose positions fluctuate over time based on the intensity and direction of the wind. As a result of phumdi proliferation and movement, the area under open water continuously fluctuates. For research based on reflectance, it becomes crucial to extract the open water surface in satellite imagery. The current study employed the Normalised Difference Water Index (NDWI) to extract the surface of open water.

Enhancement of the target items' brightness value and suppression of the background objects' brightness value can be achieved by utilising the disparity in reflectance values between the target objects in distinct bands. McFeeters built the NDWI employing the green and NIR bands based on this theory, which can efficiently suppress information about the soil and plants and boost information about water (McFeeters, 1996). The following is the equation for NDWI (Eq. 1).

$$NDWI = \frac{\lambda_{green} - \lambda_{NIR}}{\lambda_{green} + \lambda_{NIR}} \quad (1)$$

Where, λ_{NIR} indicates the reflectance of the near-infrared band (Band 4 of the ResourceSat-2 and Band 8 of the Sentinel-2 satellite imageries) and λ_{green} indicates the reflectance of the green band (Band 2 of the ResourceSat-2 and Band 3 of the Sentinel-2 satellite imageries).

2.4. Atmospheric correction for flat terrain (ATCOR)

The retrieval of reflectance in ATCOR remains constant throughout sun positions and time intervals. It enables coherent reflectance and temperature retrievals, facilitating relevant time series analysis. It makes use of absolute reflectance to enable quantitative index analysis. The support of the particular sensor is the most crucial component of atmospheric correction algorithms. The preliminary requirement of such algorithms is relative spectral response file of the particular sensor or the satellite. There are several algorithms for atmospheric corrections but their compatibility with LISS-IV sensor aboard ResourceSat-2 is limited which is one of the very important satellite/sensors in Indian Sub-continent as it provides multispectral information at very high spatial resolution. ATCOR supports the LISS-IV with the advantage of definition and addition of new sensor. ATCOR correction is intended to eliminate air impacts in order to restore the physical characteristics of the Earth's surface, which include surface temperature, soil visibility, and surface reflectance. Pre-classification of the scene, atmospheric parameter recovery, and surface reflectance recovery are the three primary features of ATCOR (Marcello et al., 2016; Richter, 1996). Eq. (2) is used to determine the surface reflectance (λ_{sup}).

$$\lambda_{sup} = \frac{1}{a_1} \left[\frac{d^2 \pi L_{TOA}}{E_{TOA} \cos \theta_i} - a_0 \right] \quad (2)$$

Where, ϕ is the solar zenith angle, d is the direct distance to the sun, L_{TOA} is the satellite's spectral radiance, and E_{TOA} is the solar spectral radiance on a surface perpendicular to the rays from the sun outside the atmosphere. It is necessary to have the standard atmospheric parameters in order to obtain the coefficients a_0 and a_1 . The adjacency effect is corrected for by computing the mean reflectance of the investigated area $\bar{\lambda}_{sup}$. Thus, the relationship to determine the reflectance of the adjacent effect's free surface is described by eq. (3).

$$\lambda'_{sup} = \lambda_{sup} + \left[\int_{a_1}^{a_2} \frac{\rho_{0dif}}{\rho_{0dir}} P da \right] \left[\lambda_{sup} - \sum_{i=1}^{n_p} \overline{\lambda_{sup}} w_i \right] \quad (3)$$

Where, w_i denotes the weighting coefficients as a function of the distance-dependence, P is the sensor-specific spectral response curve, and ρ_{0dif} and ρ_{0dir} , respectively, are the diffuse and direct transmittance.

2.5. Multiple linear regression (MLR)

MLR is a statistical method that uses a single functional formula to depict the linear relationship between a number of independent variables and a dependent variable. When it comes to understanding the majority of natural events, the dependent variable is typically influenced by more than two independent variables. This is the same premise as simple linear regression, which shows the link between one independent variable and a dependent variable. In this study, the MLR model, using one dependent variable (y) and multiple independent variables (x_i) was generated using the eq. (4):

$$y = a_1 x_1 + a_2 x_2 + a_3 x_3 + \dots + a_n x_n + c \quad (4)$$

where, a_i ($i = 1, 2, 3, \dots, k$) represents regression coefficients for the independent variables x_i ($i = 1, 2, 3, \dots, k$), c is the constant of the linear equation, and y is the dependent variable.

2.6. Support vector machine (SVM)

In the context of regression prediction, support vector regression (SVR) is an implementation of support vector machines (SVM). SVR determines a linear regression equation that can be used to compute a hyperplane that fits all sample points while minimizing the total variance of the sample points. Typically, SVR has been applied to solve nonlinear regression issues through employing a kernel function that maps high-dimensional feature space to low-dimensional sample space. The SVR fitting function for nonlinear scenarios can be expressed as eq. (5)

$$y(x) = w^T \phi(x) + b \quad (5)$$

Where, $\phi(x)$ is the mapping function, which can map low dimensional non-linear samples to high-dimensional feature space; the coefficients are w and b are estimated by minimizing the regularised function $R(C)$ (Eq. (6,7)):

$$R(C) = \frac{C}{N} \sum_{i=1}^N L_e(d_i, y_i) + \frac{1}{2} \|w\|^2 \quad (6)$$

Where,

$$L_e(d_i, y_i) = \begin{cases} |d_i - y_i| - \varepsilon & \text{if } |d_i - y_i| \geq \varepsilon \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Considering that there may be some errors in solving practical problems, two slack variables ξ_i and ξ_i^* are introduced within the allowable range of errors. Hence, the regression fitting problem can be expressed as (Eq. (8)):

$$R(C) = \frac{1}{2} \|w\|^2 + C \left[\sum_{i=1}^N (\xi_i + \xi_i^*) \right] \quad (8)$$

$$\text{Subjected to. } \begin{cases} d_i - w\phi(x_i) - b_i \leq \varepsilon + \xi_i, i = 1, 2, 3, \dots, N \\ w\phi(x_i + b_i - y_i) \leq \varepsilon + \xi_i^*, i = 1, 2, 3, \dots, N \\ \xi_i, \xi_i^* \geq 0, i = 1, 2, 3, \dots, N. \end{cases}$$

where, d_i is the intended value, and C is the regularised constant that was used to calculate the trade-off between the regularised term and the empirical error. ε is referred to as the SVM tube size.

The decision function of the SVR model under the nonlinear issue can be obtained following the final solution, in accordance with the Karush-Kuhn-Tucker (KKT) condition:

$$y(x) = \sum_{i=1}^N (\alpha_i - \alpha_i^*) K(x, x_i) + b \quad (9)$$

α_i and α_i^* are the so-called Lagrange multipliers. They satisfy the conditions $\alpha_i, \alpha_i^* \neq 0$. The input samples are transferred to the high-dimensional feature space. The nonlinear issue is then transformed into the convex quadratic programming problem below based on the Lagrange duality argument:

$$\begin{cases} \min_{\alpha \in R^{2N}} \frac{1}{2} \sum_{i=0}^N \sum_{j=0}^N (\alpha_i^* + \alpha_i) (\alpha_i^* + \alpha_i) K(x_i, x_j) + \varepsilon \sum_{i=1}^N (\alpha_i^* + \alpha_i) - \sum_{i=1}^N d_i (\alpha_i^* - \alpha_i) \\ \text{subjected to. } \sum_{i=1}^N (\alpha_i^* - \alpha_i) = 0, 0 \leq \alpha_i, \alpha_i^* \leq C, i = 1, 2, 3, \dots, N \end{cases}$$

Where $K(x_i, x_j)$ is defined as kernel function. In this study Gaussian Radial basis kernel and linear kernel were used.

$$\text{Linear kernel : } K(x_i, x_j) = x_i^T \times x_j \quad (10)$$

$$\text{Radial basis kernel : } K(x_i, x_j) = \exp(-\|x_i - x_j\|^2), \gamma > 0 \quad (11)$$

Where, d , r , and γ are kernel parameters

2.7. Random forest (RF)

Machine learning is the foundation of the Random Forest Regression (RFR) technique. According to Breiman (2001), RF is intended to provide results using the single-tree method's average and the majority vote. The bagging and random subspace approaches serve as its cornerstones. Bagging is used to generate multiple learning trees, which are then combined to get the final prediction. The learner trees are trained using several independent bootstrap samples generated from the original training set. While generating the bootstrap samples, replacement of examples is permitted. For the bootstrap samples with input vector x , F separate regression trees are constructed. Regression trees are distinguished by their high variance and low bias. In regression tasks, the random forest prediction is obtained by computing $g_f(x)$, which is the mean prediction of F regression trees (Seo et al., 2017). The objective of bagging is to prevent the ensemble RFR model from overfitting and to minimise its variance. Consequently, there can be no correlation between the learner trees (Eq. (12)).

$$\text{RF prediction} = \frac{1}{F} \sum_{f=1}^F g_f(x) \quad (12)$$

A lower Gini value is indicative of a purer node, and RF uses this indication to evaluate attribute purity (Ahmed and Hikmat Sadiq, 2018). The objective of bagging is to prevent the ensemble RFR model from overfitting and to minimise its variance. Consequently, there can be no correlation between the learner trees. An out-of-bag data (OOB) dataset is generated by compiling the samples from the original training dataset that were not chosen to train the f^{th} regression tree during bagging. The following represents the estimation eq. (13):

$$MSE_{OOB} = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y}_i)^2 \quad (14)$$

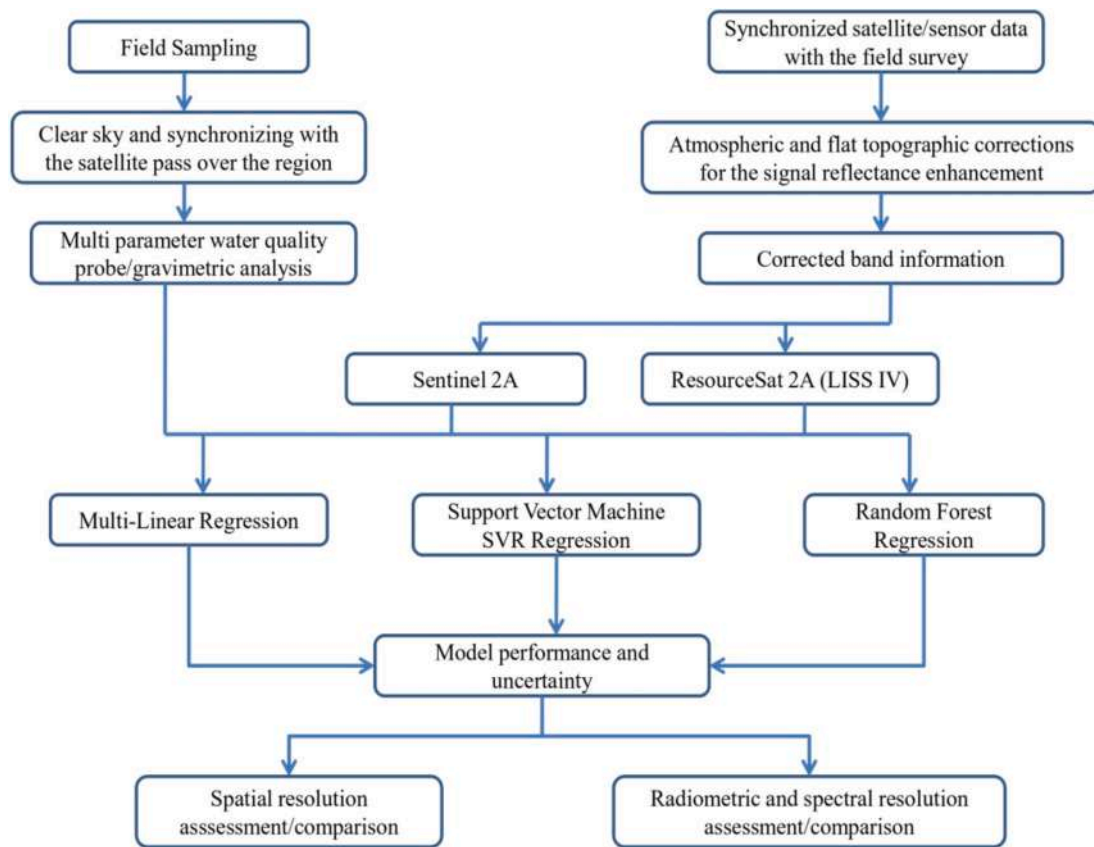


Fig. 2. Methodological hierarchy.

Where, the average prediction for the i^{th} observation from all trees for which this observation is OOB is represented by $\bar{y}_i$

2.8. Hierarchy of machine learning model development

The precise correction of atmospheric effects is essential for the accurate analysis of inland water color from remote sensing data. Hence the satellite remote sensing data were atmospherically corrected using ATCOR algorithm. In order to extract the open water surface of the lake NDWI was employed. The water quality model was developed by importing the atmospherically corrected and *in-situ* measured datasets into the Python environment. The datasets were initially normalised using the “MinMaxScaler” transformation. It scales every single attribute to a specified range in order to standardise features. This transformation scales and translates each feature individually such that it is in the given range on the training set, i.e. between zero and one (Amorim et al., 2023). Data was divided into training and testing sets in a 60:40 ratio prior to the development of the machine learning-based statistical model. Nonetheless, there are 50:50, 70:30, 80:20, 75:25 data split ratios have also been used in the several studies in the past (Joseph, 2022; Nazarkar et al., 2023). After splitting data, ML algorithms were trained and tested based on the split datasets respectively.

Regression models MLR, SVR, and RFR were employed in this research due to their past validity and versatility in a multitude of water quality modelling studies (Masood et al., 2023; Nouraki et al., 2021; Parveen et al., 2017). The random forest regression (RFR) method is widely utilised for high-dimensional data analysis (Cheng et al., 2023). RFR reduces overfitting in decision trees. However, they exhibit notable fluctuations in response to even slight modifications in the data (Shanmugasundar et al., 2021). Utilising a kernel function to project the initial data onto a high-dimensional linearly separable space provides an immense advantage in SVR when dealing with nonlinear processes

(Pilario et al., 2020).

In order to achieve higher level model accuracy, the ML technique’s hyper-parameters are adjusted (Yang and Shami, 2020). Grid search and random search techniques are the most commonly used optimization techniques for hyper-parameters (Bischl et al., 2023). However, in this study grid search was used for optimization. The model was validated based on k-fold cross validation approach. In order to evaluate model performance, statistical indices namely Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and Coefficient of determination (R^2) were used (eq. 15–18). There are several algorithms which have been used in the past to determine the uncertainty associated with the model predictions such Monte-Carlo simulation (Pakyuz-Charrier et al., 2019), percent of relative error index (PREI) (Bui et al., 2020), 95 Prediction Probability Unit (PPU) (Anand et al., 2024; Cha et al., 2023). In this study 95 PPU plot was employed. Understanding the uncertainty in model predictions is usually involved in a 95 % Prediction Probability Unit (PPU) graphic. In modelling, this type of illustration is frequently employed to assess the extent to which prediction model function. Five percent of the weaker forecasts are rejected when the value is computed between 2.5 % and 97.5 % of the output variable. The 95 % prediction uncertainties are computed by (eq. 19). Fig. 2 shows the methodological hierarchy.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n |y - y_i|^2} \quad (15)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y - y_i| \quad (16)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|y - y_i|}{y} \cdot 100\% \quad (17)$$

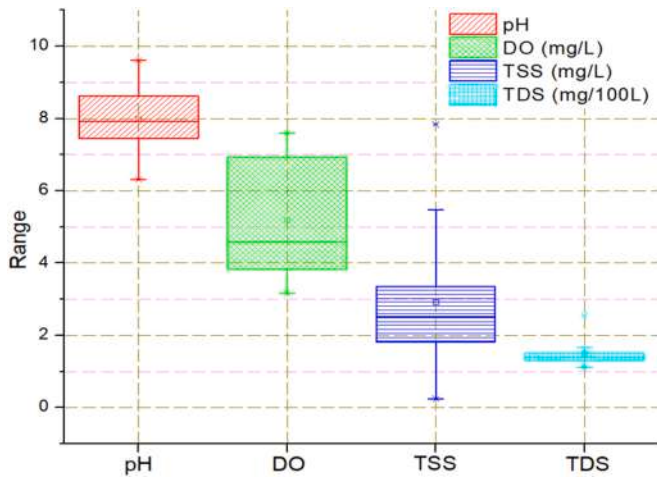


Fig. 3. Statistical attributes of various water quality parameters.

Table 2

The empirical algorithms generated using Multi-Linear Regression technique for Sentinel and ResourceSat

Satellite	Parameter	Equations	RMSE, R ²
Sentinel	DO	DO = 30.662 – (0.016 x B2) – (0.009 x B3) + (0.016 x B4) + (0.001 x B8)	0.42, 0.81
	pH	pH = 23.389 – (0.002 x B2) – (0.011 x B3) + (0.006 x B4) + (4.2 × 10 ⁻⁵ x B8)	0.13, 0.78
	TDS	TDS = -7.056 + (0.112 x B2) + (0.126 x B3) – (0.227 x B4) + (0.035 x B8)	0.52, 0.75
	TSS	TSS = 19.59 – (0.015 x B2) + (0.005 x B3) + (0.004 x B4) + (3.6 × 10 ⁻⁵ x B8)	0.46, 0.71
ResourceSat	DO	DO = 8.09 – (0.011 x B4) + (0.072 x B3) – (0.102 x B2)	2.65, 0.74
	pH	pH = 5.62 – (0.045 x B4) – (0.119 x B3) + (0.139 x B2)	0.39, 0.82
	TDS	TDS = 174.70 – (0.576 x B4) + (3.837 x B3) – (2.599 x B2)	0.76, 0.73
	TSS	TSS = 2.974 + (0.051 x B4) + (0.028 x B3) – (0.071 x B2)	0.54, 0.69

$$R^2 = 1 - \frac{\sum_{i=1}^n (y - y_i)^2}{\sum_{i=1}^n (y - \bar{y})^2} \quad (18)$$

$$\text{Bracketed in 95 PPU} = \frac{1}{n} \text{count} (N | X_L \leq N \leq X_U) \times 100 \quad (19)$$

Where, y represents the observed values and y_i represents the predicted value resulting from regression model, n represents number of observations, N increases with observations falling between the corresponding X_L and X_U increase and “Bracketed in 95 PPU” denotes the number of observed data bracketed by a 95 % confidence interval (Abbaspour et al., 2007; Cha et al., 2023; Ebtehaj et al., 2020)

3. Results and discussions

3.1. Field sampling and model performance assessment

Statistical attributes of various water quality parameters in the Loktak Lake is shown in Fig. 3. The data was compiled considering the seasonal variations in the Loktak Lake. The pH was found in the range of 5.5 to 9 with an average variation of ± 1.5 , dissolved oxygen in the range 4 to 9.5 mg/l with an average variation of ± 3 mg/l, total dissolved solids in the range 110 to 160 mg/l with an average variation of ± 20 mg/l, and

Table 3

Machine learning models performance indices for Sentinel.

Regression Technique	Parameters	Statistical Indices
Random Forest	DO	Test- R ² = 0.97, RMSE = 0.24, MAE = 0.14, MAPE = 3.1 % Train- R ² = 0.98, RMSE = 0.16, MAE = 0.17, MAPE = 3.3 %
	pH	Test- R ² = 0.96, RMSE = 0.18, MAE = 0.25, MAPE = 3.4 % Train- R ² = 0.97, RMSE = 0.25, MAE = 0.31, MAPE = 3.7 %
	TDS	Train- R ² = 0.97, RMSE = 6, MAE = 3.14, MAPE = 2.1 % Test- R ² = 0.97, RMSE = 7.3, MAE = 3.3, MAPE = 2.4 %
	TSS	Train- R ² = 0.84, RMSE = 0.24, MAE = 0.22, MAPE = 13.7 % Test- R ² = 0.85, RMSE = 0.31, MAE = 0.26, MAPE = 14.5 %
Support Vector Machine	DO	Test- R ² = 0.99, RMSE = 0.07, MAE = 0.049, MAPE = 1.18 % Train- R ² = 0.99, RMSE = 0.11, MAE = 0.056, MAPE = 1.39 %
	pH	Test- R ² = 0.99, RMSE = 0.05, MAE = 0.08, MAPE = 1.2 % Train- R ² = 0.99, RMSE = 0.07, MAE = 0.11, MAPE = 1.38 %
	TDS	Test- R ² = 0.99, RMSE = 0.1, MAE = 0.055, MAPE = 0.01 % Train- R ² = 0.99, RMSE = 0.13, MAE = 0.061, MAPE = 0.01 %
	TSS	Test- R ² = 0.78, RMSE = 0.28, MAE = 0.24, MAPE = 10.9 % Train- R ² = 0.80, RMSE = 0.34, MAE = 0.29, MAPE = 12.1 %

Table 4

Machine learning models performance indices for ResourceSat.

Regression Technique	Parameters	Statistical Indices
Random Forest	DO	Test- R ² = 0.98, RMSE = 0.16, MAE = 0.13, MAPE = 2.8 % Train- R ² = 0.99, RMSE = 0.20, MAE = 0.16, MAPE = 3.4 %
	pH	Test- R ² = 0.96, RMSE = 0.22, MAE = 0.17, MAPE = 2.2 % Train- R ² = 0.98, RMSE = 0.27, MAE = 0.23, MAPE = 2.7 %
	TDS	Test- R ² = 0.94, RMSE = 6.22, MAE = 3.82, MAPE = 2.8 % Train- R ² = 0.96, RMSE = 5.37, MAE = 3.45, MAPE = 3.5 %
	TSS	Test- R ² = 0.79, RMSE = 0.2, MAE = 0.17, MAPE = 9.8 % Train- R ² = 0.80, RMSE = 0.28, MAE = 0.23, MAPE = 10.5 %
Support Vector Machine	DO	Test- R ² = 0.99, RMSE = 0.08, MAE = 0.08, MAPE = 1.7 % Train- R ² = 0.99, RMSE = 0.12, MAE = 0.11, MAPE = 2.1 %
	pH	Test- R ² = 0.99, RMSE = 0.11, MAE = 0.09, MAPE = 1.2 % Train- R ² = 0.99, RMSE = 0.16, MAE = 0.12, MAPE = 1.4 %
	TDS	Test- R ² = 0.99, RMSE = 0.1, MAE = 0.08, MAPE = 0.1 % Train- R ² = 0.99, RMSE = 0.13, MAE = 0.1, MAPE = 0.1 %
	TSS	Test- R ² = 0.96, RMSE = 0.13, MAE = 0.1, MAPE = 5.1 % Train- R ² = 0.97, RMSE = 0.21, MAE = 0.14, MAPE = 5.6 %

total suspended solids in the range of 1.5 to 5 mg/l with an average variation of ± 2.5 mg/l. The changes in the pH, dissolved oxygen, total dissolved solids and total suspended solids are mainly due to drastic fluctuations in the temperature, precipitation and streamflow contributing to the lake during pre-monsoon and post-monsoon seasons. Anthropogenic activities such as agriculture practices, amount of effluent brought by the intersecting streams can influence the pH and TDS in the lakes hence affecting the dissolved oxygen. Similar type of trends was observed in the Loktak Lake in the studies carried out by (Laishram et al., 2022; Mayanglambam and Neelam, 2020; Tuboi et al., 2018)

In this study satellite imageries from two different sensors were tested using three different techniques. Among the two sensors used in this study, the Sentinel multispectral images had comparatively low spatial resolution and high radiometric resolution, while ResourceSat had high spatial resolution and low radiometric resolution. The accuracy of the two sensors for each of the three models—MLR, RFR, and SVR—is displayed in Tables 2, 3, and 4. Irrespective of the sensor, Support Vector Regression yielded the highest accuracy followed by the Random Forest

Regression and Multi-Linear Regression. The water quality parameters model was designed to forecast the quality of the water by using the reflectance of the water that the sensors have detected. The optical bands were used in this study as independent parameters for model development as they highly sensitive to changes in the water quality (Leggesse et al., 2023; Villota-González et al., 2023; Zhu et al., 2022). Four Sentinel bands and three ResourceSat bands were used as input variables for the model simulations in order to estimate pH, total suspended particles, total dissolved solids, and dissolved oxygen. The Support Vector Regression for Sentinel-2 had the lowest root mean square error, ranging from 0.05 to 0.28; the Random Forest Regression, with total dissolved solids being the lone exception, came closely behind, at 0.18 to 0.24; and the Multi-Linear Regression, with a range of 0.13 to 0.52 was the highest. The Support Vector Regression model produced a better result in the ResourceSat-2 case, with the root mean square ranging between 0.08 and 0.13. The Random Forest Regression model followed with values ranging between 0.16 and 0.22 (the total dissolved solids being the only exception), and Multi-Linear Regression model produced poor results (0.39–2.65). However, to quantify the

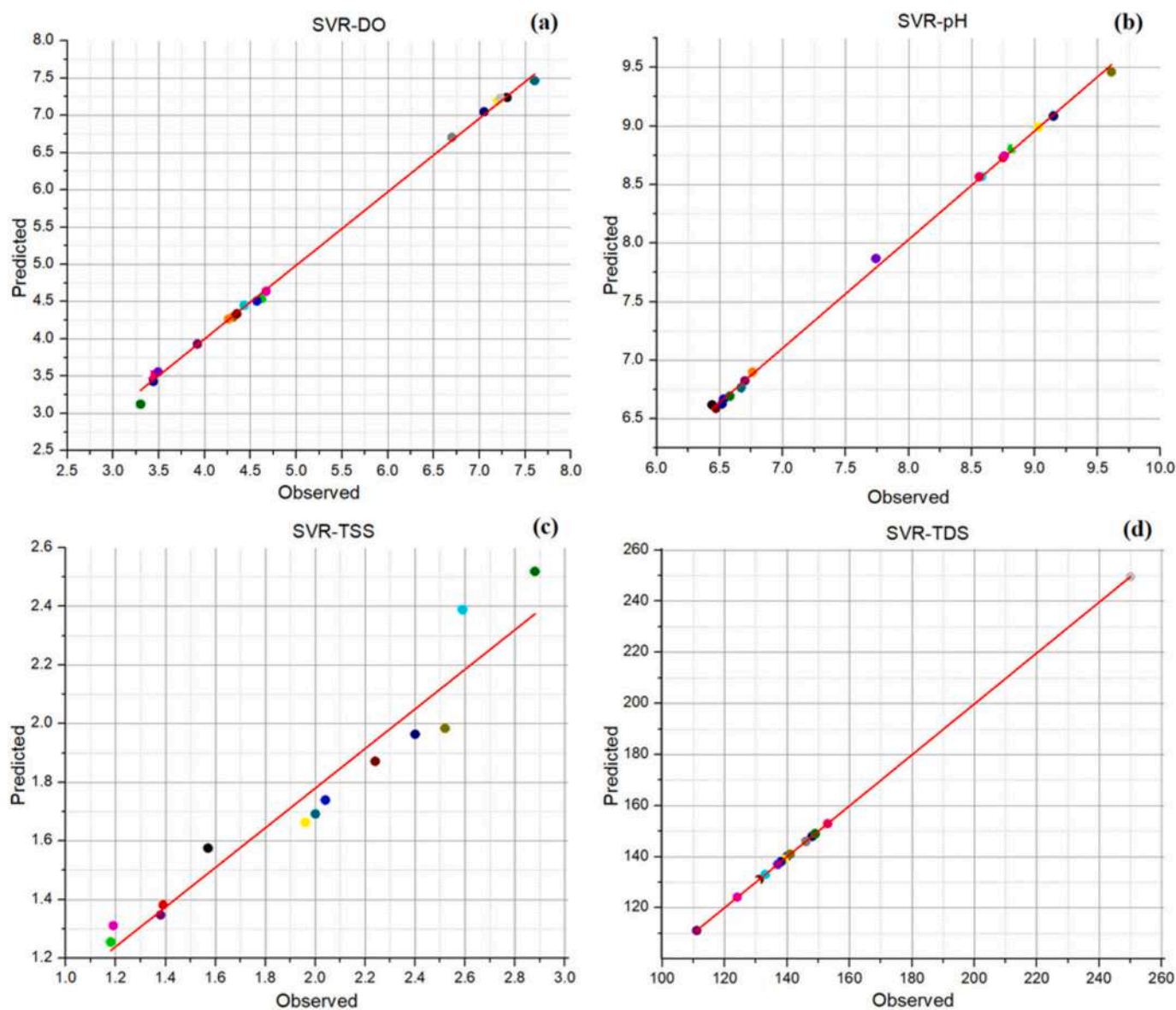


Fig. 4. Observed versus predicted values using Support Vector Machine and Random Forest models generated from Sentinel multi-spectral satellite imageries. Support Vector Machine models: (a) Dissolved oxygen, (b) pH, (c) Total Suspended Solids, (d) Total Dissolved Solids. Random Forest models: (e) Dissolved oxygen, (f) pH, (g) Total Suspended Solids, (h) Total Dissolved Solids.

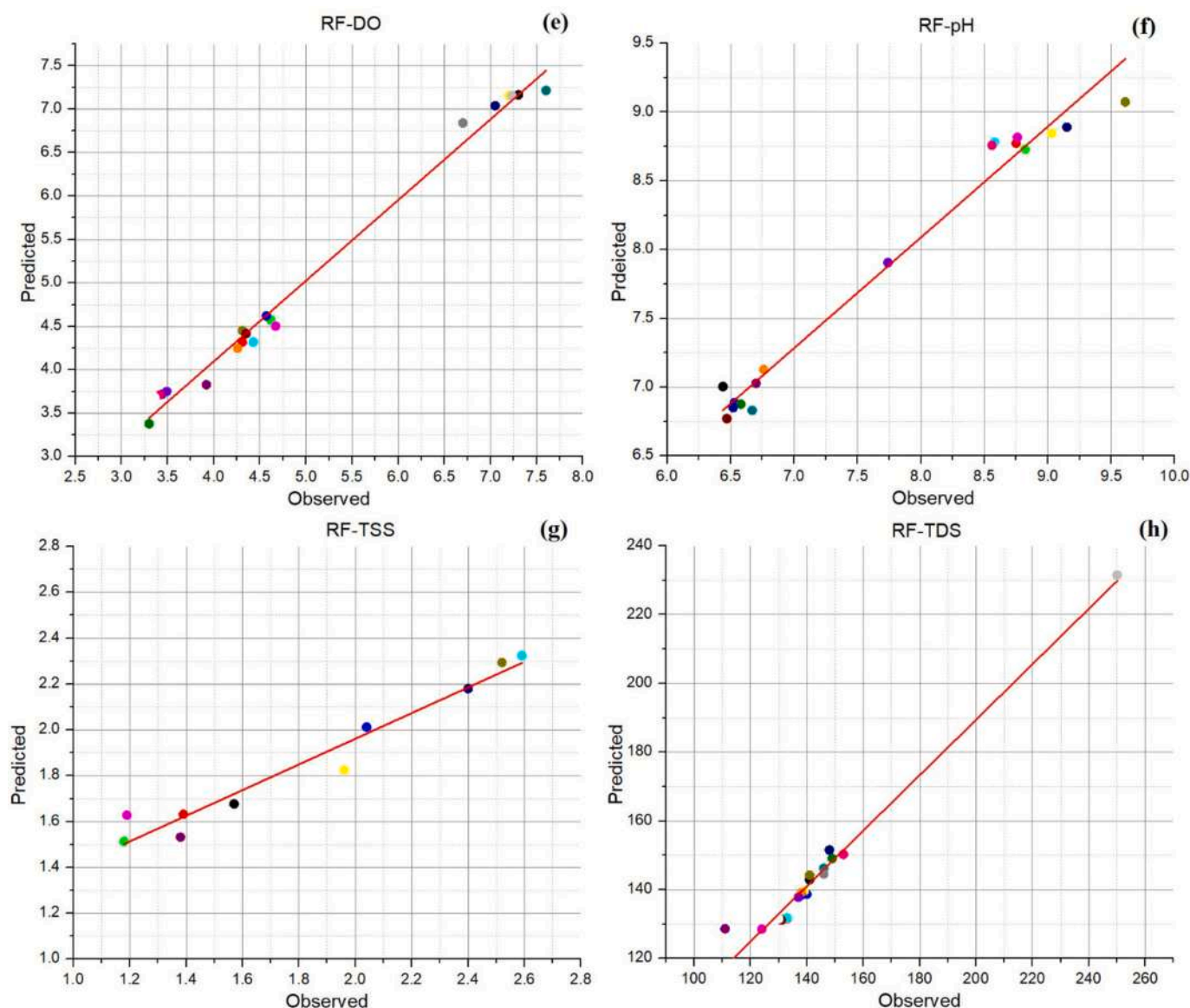


Fig. 4. (continued).

robustness of model performance two more indices were added for RF and SVR interpretation (Tables 3,4). The main reason behind the better performance of SVM model could be, the capacity of the SVM model to maximise the margin between data points and the decision frontier, which makes it especially well-suited for high-dimensional spaces, could be one of the main causes of its superior performance when compared to the RF model. It is able to generate intricate limits on decisions by utilising kernel tricks. SVM can be more effective when working with sparse data because it only uses a portion of the support vectors to generate predictions, which lowers complexity. Tables 3 and 4 lists the model performance indices for both sensors under various conditions. The linear regression models for pH, total suspended particles, total dissolved solids, and dissolved oxygen that were created using Sentinel-2 and ResourceSat-2 are highlighted in Table 2.

For Sentinel, the blue and the red spectral bands were the most sensitive independent variable followed by green band in the visible spectrum, while the near-infrared band showed the weakest correlation. With the exception of the blue spectral band, which ResourceSat lacked, identical patterns were noted. Figs. 4 and 5 for Sentinel and ResourceSat, respectively, display the prediction correlation between the predicted and observed values for each of the water quality metrics

determined in this study. When compared to the other water quality metrics, the total suspended solids demonstrated the highest level of variance, regardless of the three methods used in this study to estimate the parameters based on reflectance. Multi-Linear Regression was outperformed by the two machine learning methods, Support Vector Regression and Random Forest Regression. The mapping of water quality indicators in Loktak Lake using Sentinel and ResourceSat satellite imagery was done using Support Vector Regression models, which were chosen based on model performance.

The uncertainty in the model was analysed using the 95 PPU. From the model uncertainty analysis, it can be observed that support vector machine performed better than the random forest. The majority of the data fell within the 95 PPU plot thickness. If 95 % of the observed data points fall within the intervals, it indicates the model adequately captures the variability in the data. This thickness of the intervals reflects the uncertainty in the predictions. However, in case of SVM the thickness of the intervals was narrower as compared to RF, where the interval was found to be thicker. Narrow intervals indicate less uncertainty in the predictions and the majority of data points fall within narrow intervals, which suggests the model is highly reliable. Figs. 6 and 7 represents the 95 PPU plot for RF and SVM models for Sentinel multispectral imageries

respectively. The uncertainty of RF and SVM models for ResourceSat (LISS IV) multispectral imageries is shown in the Fig. 8 and Fig. 9 respectively.

3.2. Spatio-temporal variability of water quality parameters

Spatial variation of water quality for April (dry season) and November (wet season) for Sentinel is shown in Fig. 10 and Fig. 11 respectively. Fig. 12 shows the spatial distribution of water quality parameters for the ResourceSat for the month of March. Investigating the viability of using Sentinel and ResourceSat to retrieve water quality indicators from the data-scarce Loktak Lake was an objective of this research. Water quality characteristics could be derived from spectral reflectance data, pursuant to the developed models, which yielded higher accuracy results. Using machine learning models, specifically Support Vector Machine and Random Forest, this study was able to acquire more accuracy than it accomplished with conventional linear regression models. From the obtained results two comparisons were made: (1) seasonal variations in the water quality parameters; (2) mean

variation in the water quality parameters between two different sensors. In order to check the seasonal variations in water quality parameters, different models for various parameters generated from Sentinel was chosen. Water quality extracted from Sentinel multispectral satellite imageries was compared for the month of April (dry season) and November (wet season). The mean values for pH, dissolved oxygen, total suspended solids, and total dissolved solids were found to be 7.51, 5.72 mg/l, 2.41 mg/l and 148.58 mg/l respectively during the month of April. While, during the month of November the mean value of pH, dissolved oxygen, total suspended solids, and total dissolved solids were found to be 7.78, 8.96 mg/l, 5.10 mg/l and 127.19 mg/l respectively. The similar kind of quantitative trends in the water quality parameter were obtained in the research studies carried out by Mayanglambam and Neelam (2020) and Laishram et al. (2022). This spatio-temporal variations in the Loktak Lake is caused due to drastic shift in heat and wind fluxes South-West Indian monsoon season. Due to excessive heating of the water in the lake caused by the heat fluxes during the summer season increases the water temperature which leads to in decrease the pH and dissolved oxygen in the lake. The total suspended solids during the post

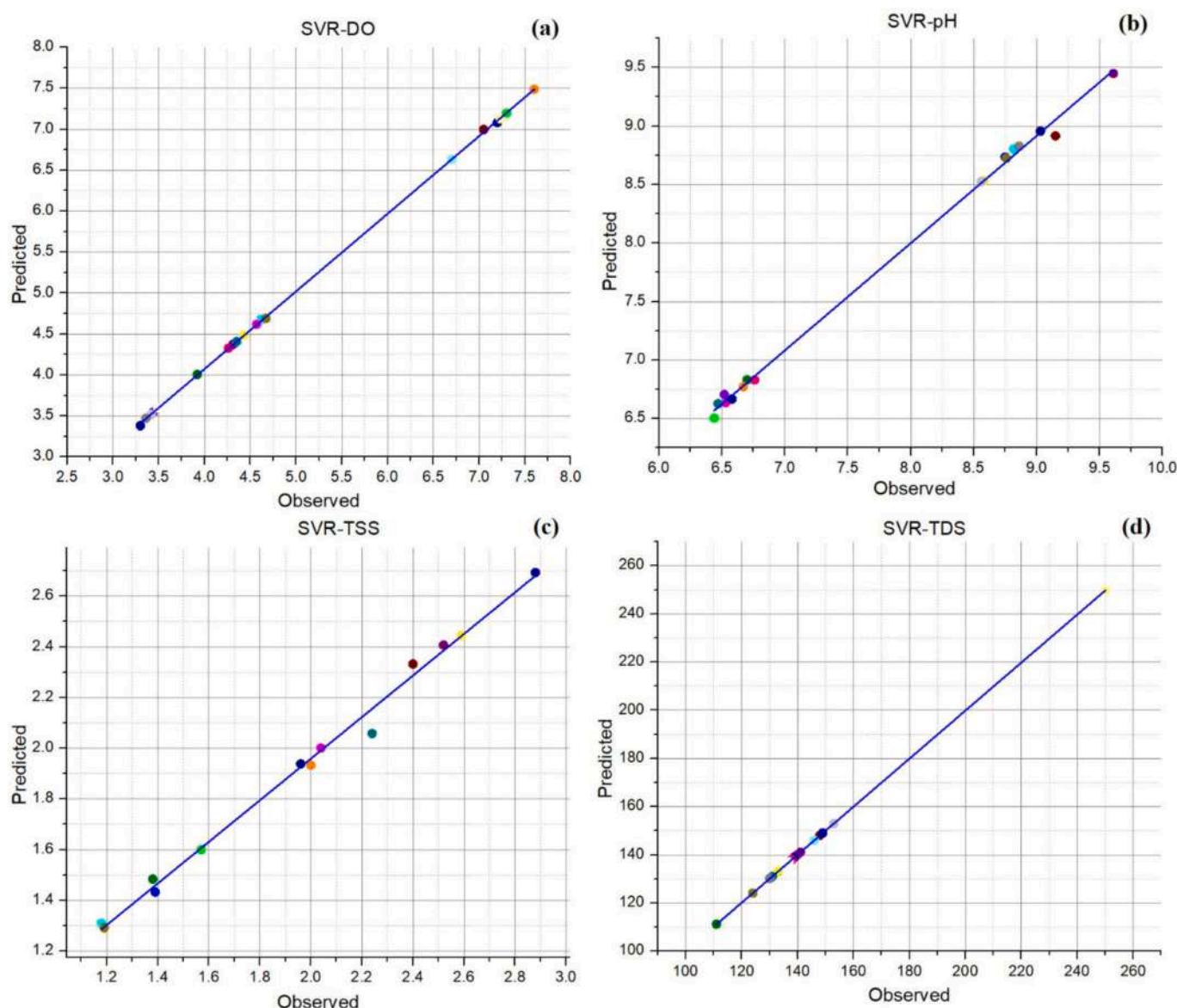


Fig. 5. Observed versus predicted values using Support Vector Machine and Random Forest models generated from ResourceSat multi-spectral satellite imageries. Support Vector Machine models: (a) Dissolved oxygen, (b) pH, (c) Total Suspended Solids, (d) Total Dissolved Solids. Random Forest models: (e) Dissolved oxygen, (f) pH, (g) Total Suspended Solids, (h) Total Dissolved Solids.

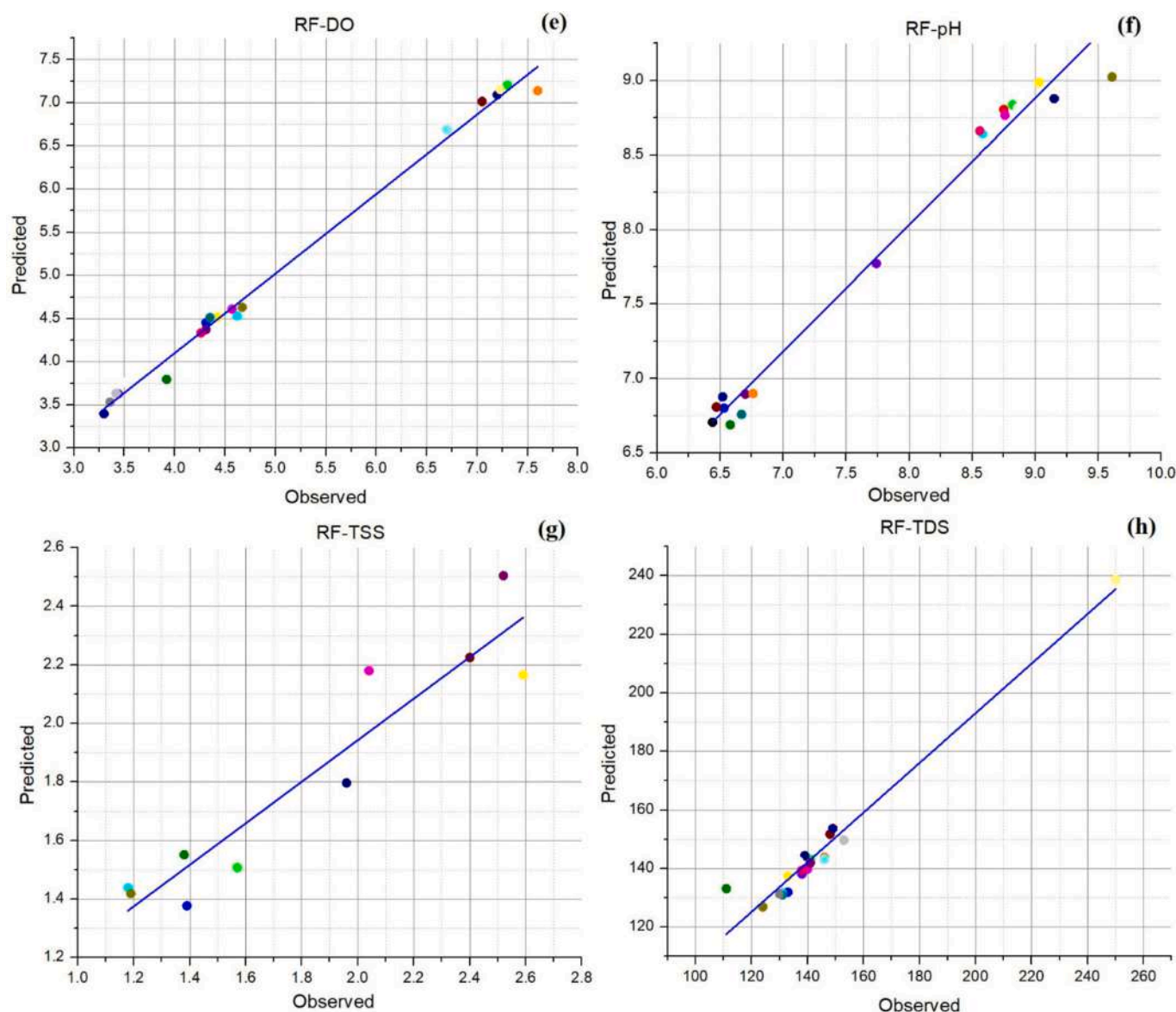


Fig. 5. (continued).

monsoon season in the month of November was much higher as compared to the pre-monsoon due to sediments driven by the streams intersecting the lake from the upstream mountainous reaches of the Himalayas. An average zonal variability statistical comparison was made with gauged data. From zonal statistical variability it was observed that deviation in the water quality parameters predicted from both the sensors/satellites based on machine learning model as compared gauged data was comparatively higher in on-shore predictions as compared to the off-shore predictions. An on-shore and off-shore comparison of both the sensors is shown in Fig. 13.

3.3. Inter-comparison of sensors/satellites and their effects of resolutions

For the evaluation of mean variation in the water quality parameters between two different sensors, the multi-spectral satellite imageries on 9 March 2022 was chosen, since this was the only date which intersected between *in-situ* water quality sample collection, pass of Sentinel and ResourceSat satellite over the Loktak Lake. In case of water quality model generated from Sentinel satellite imageries returned a mean value of 7.08, 4.95 mg/l, 1.81 mg/l and 150.85 mg/l for pH, dissolved oxygen, total suspended solids, and total dissolved solids respectively. Whereas,

in case of the models generated from ResourceSat, pH, dissolved oxygen, total suspended solids, and total dissolved solids was found to be 7.15, 4.46 mg/l, 1.74 mg/l and 150.41 mg/l. The mean values obtained from both the satellites showed very similar kind of results.

Although ResourceSat (LISS-IV) has finer spatial resolution than Sentinel-2, the later satellite has more spectral bands. In this study, the three bands present in the visible spectrum and a NIR bands were used from the Sentinel-2, where in ResourceSat two bands present in the visible spectrum and a NIR bands were used. As depicted from Fig. 11 and Fig. 12, sharper variations in water quality can be observed in the parameters extracted from the Sentinel as compared to that of ResourceSat. The main reason behind the detection of sharper variation in Sentinel can be higher spectral resolution in terms of blue band and higher radiometric resolution compared to the ResourceSat (LISS IV). Since the ResourceSat (LISS IV) have higher spatial resolution, the lower radiometric and spectral resolutions were traded off. Since the radiometric resolution of the ResourceSat (LISS IV) is less, the capacity to detect the minor fluctuations in the amount of energy decreases. An instantaneous field of view (IFOV) of the sensor must be short in order to achieve high spatial resolution. But when the area of the ground resolution cell within the IFOV gets smaller, this decreases the amount of

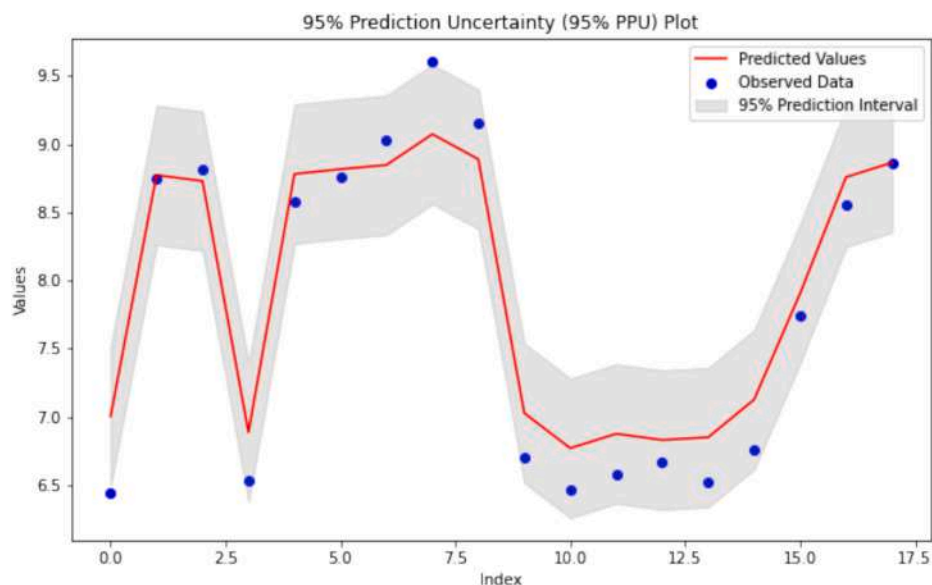


Fig. 6. 95 PPU uncertainty plot of Random Forest model generated from Sentinel.

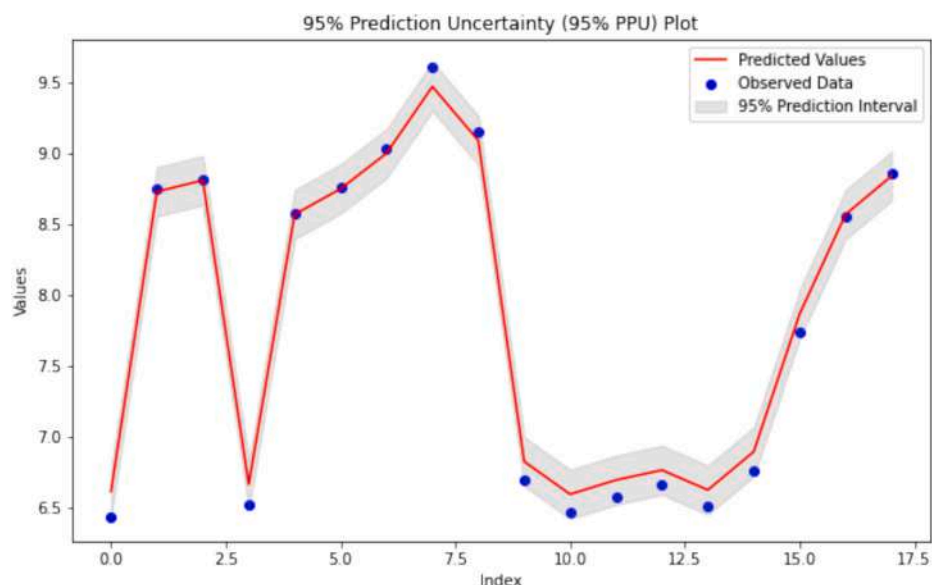


Fig. 7. 95 PPU uncertainty plot of Support Vector Machine model generated from Sentinel.

energy that can be detected. This results in a decrease in radiometric resolution, or the capacity to identify minute variations in energy. Broadening the wavelength range detected for a specific band is essential in order to increase the radiometric resolution without reducing the spatial resolution. However, this lowers the sensor's spectral resolution. In contrast, better spectral and radiometric resolution would be possible with coarser spatial resolution. However, there are benefits associated with having a higher spatial resolution. Compared to coarser-resolution satellites, ResourceSat's higher spatial resolution may make it possible to remotely sense the water quality in smaller ponds, reservoirs, and narrow-width streams. Since many streams with limited width and man-made aqua ponds for fishing dominate the Loktak Lake catchment. Due to their small size, they are challenging to monitor with Landsat-8 or Sentinel-2, or even ground-based sampling. For instance, since Landsat 8 and Sentinel 2 have spatial resolutions of 30 and 10 m, respectively, a stream with a width of 30 m would be covered by either one Landsat-8 pixel or three Sentinel-2 pixels. A ResourceSat (LISS-IV) image would contain about 6 pixels for a stream with a width of 30 m. It can therefore

offer superior spatial information.

The outcome of this study clearly shows the superiority of machine learning techniques over the other conventional techniques. The results of this study, which employed machine learning approaches, are consistent with the research conducted by Prasad et al. (2020), who discovered 94 % accuracy when mapping prospective groundwater in India using random forests and machine learning models. Abdi (2020) proposed that the reason machine learning models perform better overall with remote sensing data than parametric models is that they render it likely to uncover higher-level and non-linear statistical relations. However, a well-formulated parametric model can also yield sufficient accuracy for water quality retrieval (Abdelmalik, 2018). The results of this study clearly indicate the better performance of the SVM model over the other two models used in this study which is similar to the findings of the several other studies (Kibitia et al., 2020; İskenderoğlu et al., 2020). In comparison with the satellite remote sensing-based study carried globally, Zhu et al. (2022) obtained 1.5 %, 0.02 % and 1.7 % errors associated turbidity, dissolved oxygen and chlorophyll

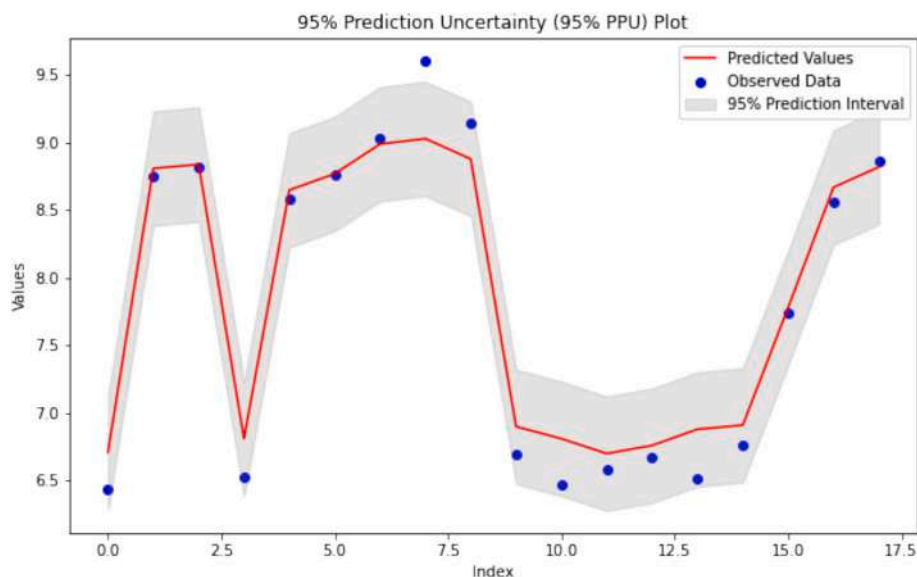


Fig. 8. 95 PPU uncertainty plot of Random Forest model generated from ResourceSat (LISS-IV).

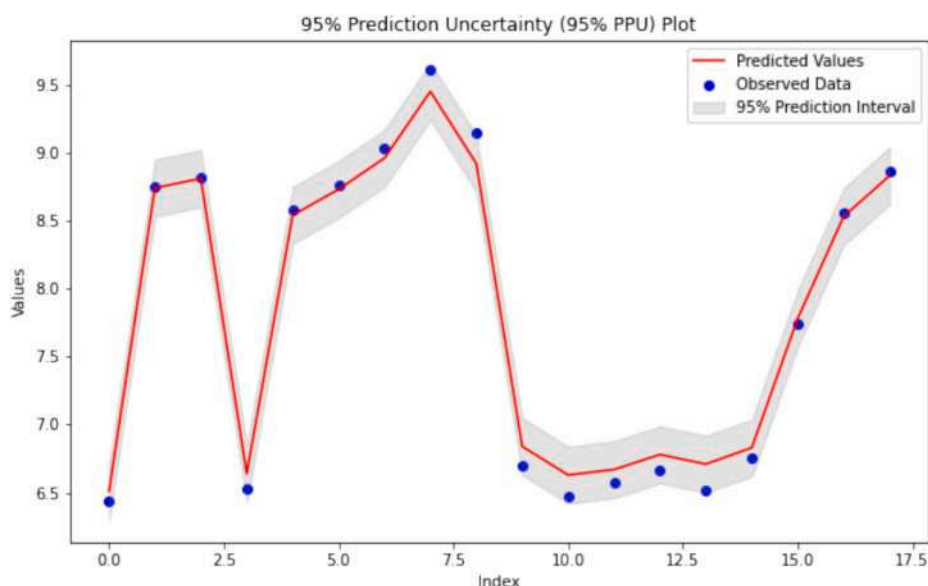


Fig. 9. 95 PPU uncertainty plot of Support Vector Machine model generated from ResourceSat (LISS-IV).

respectively while using Sentinel satellite imageries. While using Landsat satellite imageries for the retrieval of water quality parameters based on remote sensing, Jakovljevic et al. (2024) obtained an R-squared value of 0.998, 0.999, 0.990 for the chlorophyll, dissolved oxygen and total suspended solids respectively. A high R-squared value indicates that the model can adequately explain the fluctuation in the parameter using the input features while applied to an optically inactive water quality parameter that was produced through machine learning parameters. This implies that the attributes selected capture underlying patterns or relationships that contribute to the variance in the optical properties of water, even though the parameter may not have a direct impact on those qualities. The possible ramifications for water quality monitoring must be carefully considered when interpreting these outcomes.

Other approaches, that include deep learning, have been found to deliver higher estimates of water quality retrieval in the United States (Sagan et al., 2020). There are several limitations associated with this study. Primarily, while working with remote sensing satellite datasets,

trade-offs among spectral, spatial, and radiometric resolutions limit the spatial resolution of the space-borne images to medium or coarse resolutions (Gege and Dekker, 2020). However, multispectral remote sensing is a better choice for studying the spatiotemporal variation of water quality parameters in large-scale inland areas. Nonetheless, previous research has shown that a wide bandwidth makes it difficult to separate the optical characteristics of various components of a water body with complex optical properties (Chen and Shen, 2016); however, the development and application of hyperspectral sensors is an important approach to resolving this problem. Secondly, the model performance was found to be good; however there were some limitations associated with the model uncertainty especially in the onshore zones as compared to the offshore zones. This uncertainty may be because of the onshore being more vulnerable and sensitive to the anthropogenic changes. Hence, in order to enhance the model performance there is a need to integrate the *in-situ* data number at a higher frequency from different seasons for the better capture of the variability associated with water quality parameters both spatially and temporally.

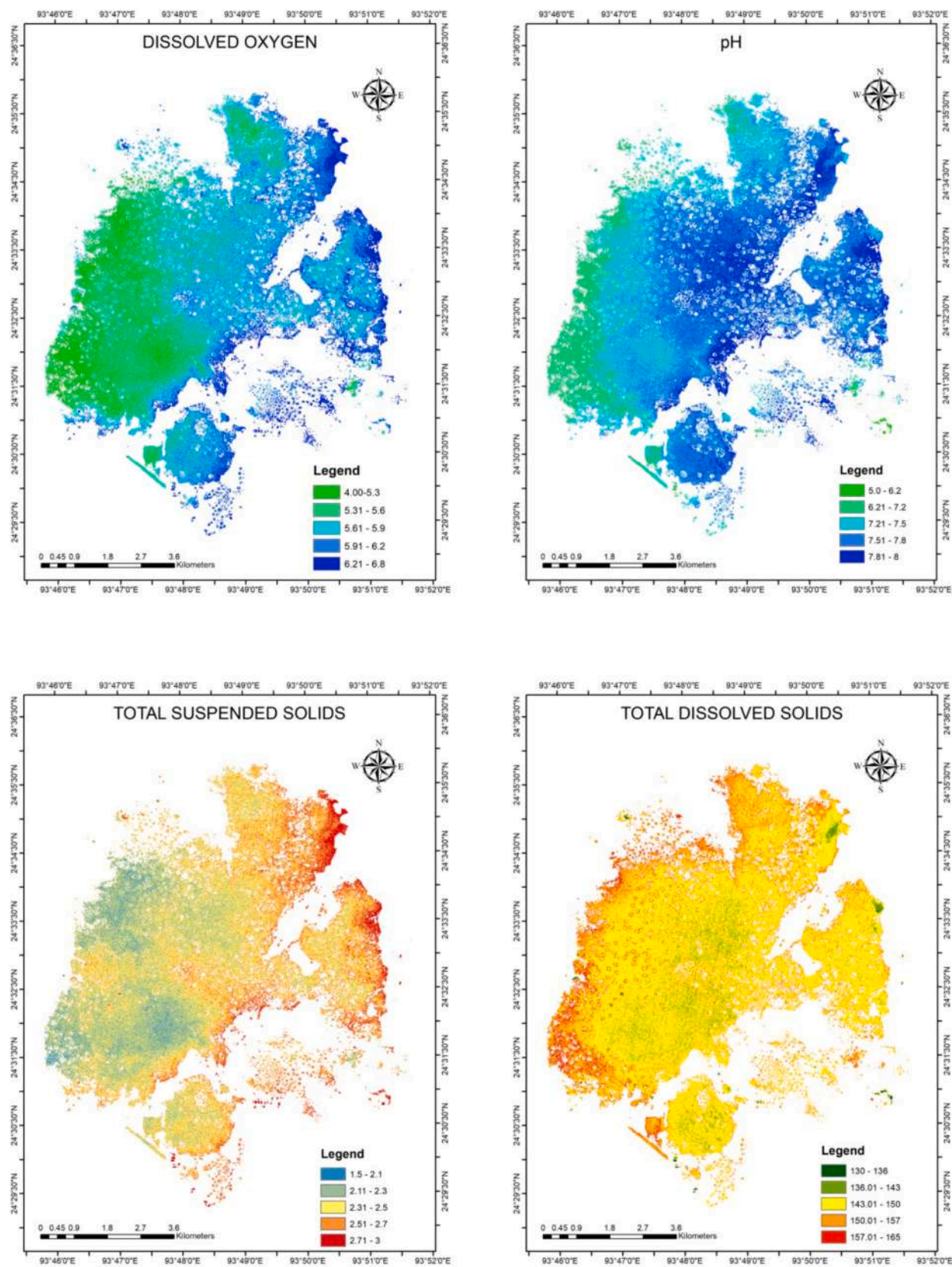


Fig. 10. Spatial distribution of water quality parameters derived from Sentinel multi-spectral satellite imagery for the month of April 2022.

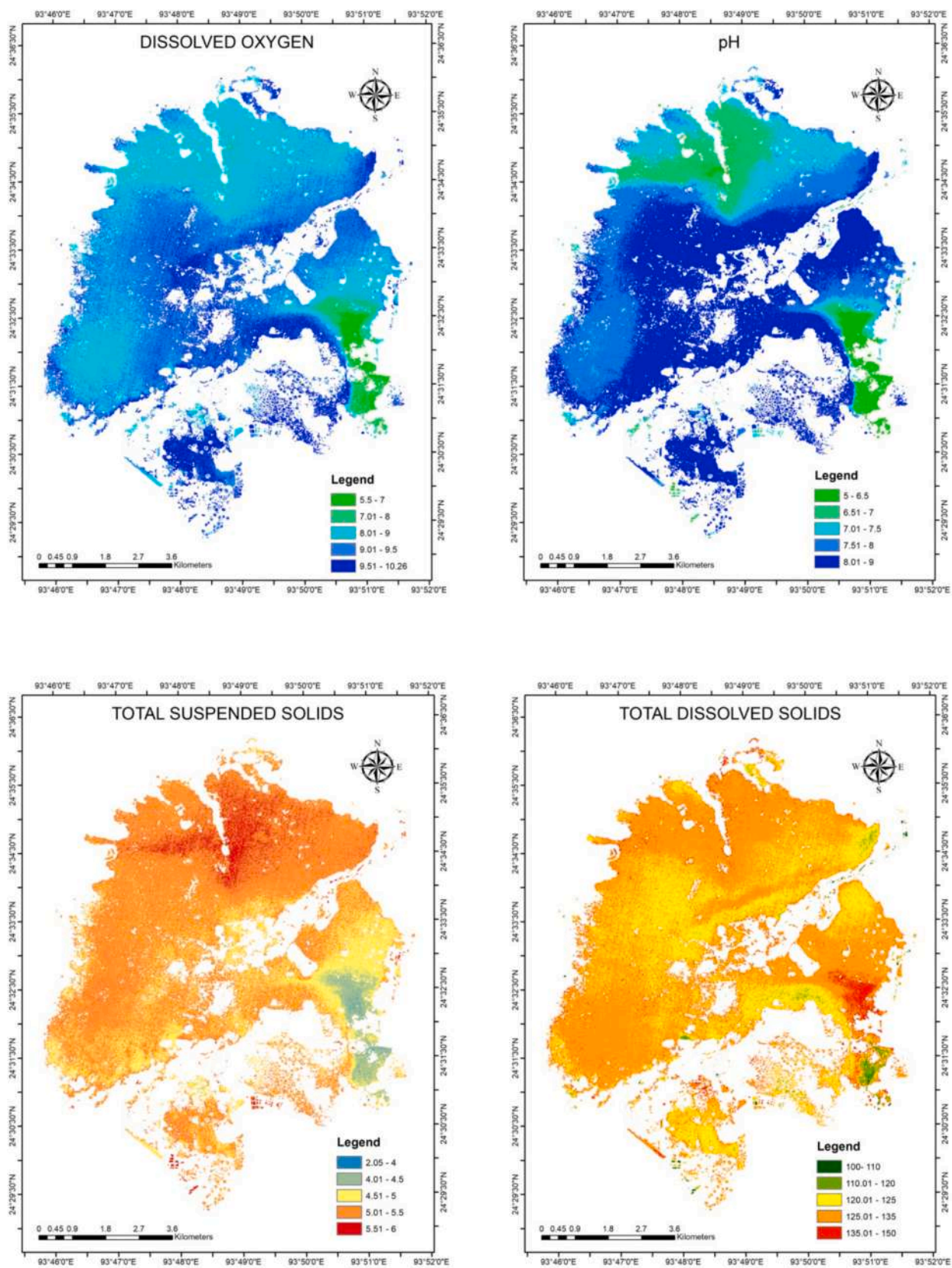


Fig. 11. Spatial distribution of water quality parameters derived from Sentinel multi-spectral satellite imageries for the month of November 2022.

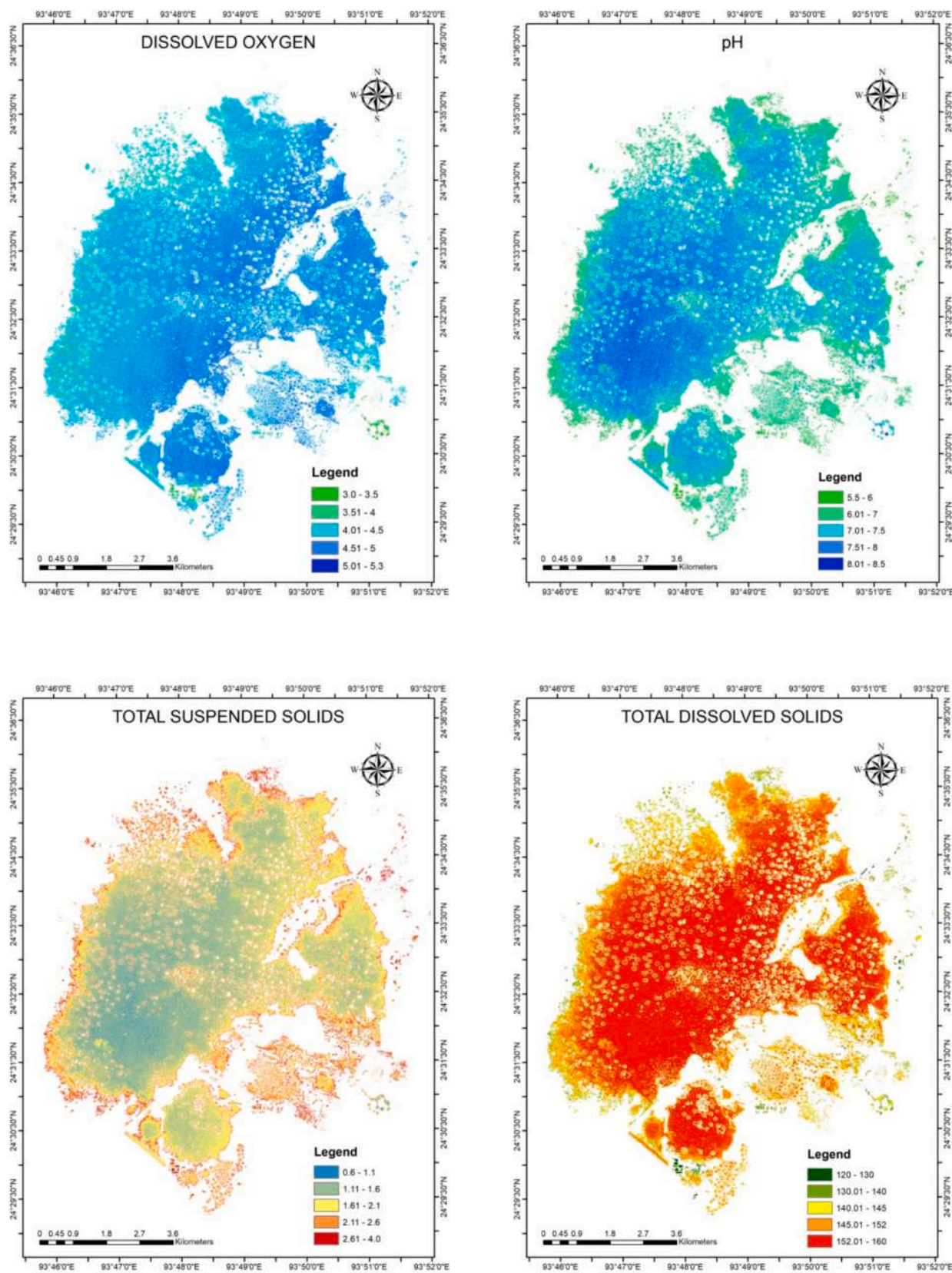


Fig. 12. Spatial distribution of water quality parameters derived from ResourceSat multi-spectral satellite imagery for the month of March 2022.

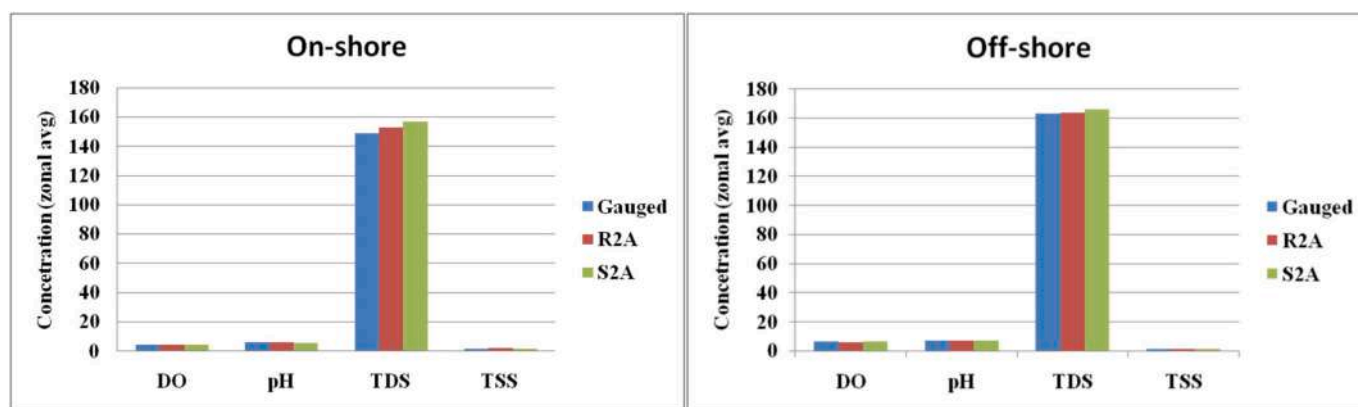


Fig. 13. Comparison between predicted and actual values for different water quality parameter from both the sensors.

4. Conclusion

Conventional water quality monitoring is restricted to pre-determined sampling sites and sporadic field sampling visits. These restrictions leave gaps in space and time when deterioration of water quality could happen and expose humans to risks. This study investigated the use of multispectral satellites imageries to monitor water quality in both temporal and spatial aspect separately. In the present study, a dynamic method for the prediction of the water quality parameters based on Sentinel-2 and ResourceSat-2 (LISS IV) in the Loktak Lake was generated and analysed. This study contrasted ResourceSat-2 (LISS IV), which has superior spatial resolution, with Sentinel-2, which has superior spectral and radiometric resolution, highlighting the benefits and drawbacks of satellite resolutions. The spectral and radiometric information obtained from these sensors were used to develop the algorithms for various water quality estimations. In order to develop the algorithms two machine learning techniques (Support Vector Machine and Random Forests), along with a conventional technique (Multi-Linear Regression) were applied. In terms of the modelling techniques, the results of machine learning techniques yielded more promising result than the conventional linear regression. There were three critical observations made on the outcomes of this research:

- The performance of Support Vector Machine was found to be superior followed by Random Forest and Multi-Linear Regression.
- The water quality model generated with Sentinel satellite imagery returned a mean value of 7.08, 4.95 mg/l, 1.81 mg/l, and 150.85 mg/l for pH, dissolved oxygen, total suspended particles, and total dissolved solids, respectively, on the identical day that both satellites passed over Loktak Lake. On the other hand, pH, dissolved oxygen, total suspended particles, and total dissolved solids were determined to be 7.15, 4.46 mg/l, 1.74 mg/l, and 150.41 mg/l for the models generated with ResourceSat.
- Water quality metrics estimates using both sensors are typically encouraging. For the assessment of water quality parameters in Loktak Lake, Sentinel-2 was found to be marginally better than ResourceSat-2 (LISS-IV) because to its better spectral and radiometric resolution when comparing the spatial distributions of observations for water quality indicators. Nonetheless, due to its superior spatial resolution, the ResourceSat-2 (LISS-IV) offers unique advantages when deployed on smaller/narrower water bodies.

The performance of the water quality algorithms that were developed using the data from both sensors was highly compelling. These results suggest that along with machine learning approach, the Sentinel and ResourceSat should be used in tandem for monitoring water quality, as they complement their respective design efforts. ResourceSat is more appropriate for assessing water quality parameters near water bodies

where fine-scale detection is necessary, such as in small water bodies, reservoirs, and narrow streams, when pixels from satellites with coarser spatial resolution are insufficient. Overall, the findings demonstrate the strong value of machine learning models in conjunction with multi-spectral satellite imagery for managing water quality, providing important assistance to decision and policymakers. The outcomes of this study can contribute in achieving the sustainable development goals (SDG) in enabling evidence-based policy and decision making in control of effluent inflow to the lake from the intersecting streams, protecting the water ecosystems, contributing to environment by monitoring quality of the waterbodies due to impact of climate and anthropogenic changes. This study has certain drawbacks besides to its numerous advantages. Since this study focuses on three algorithms and two multi-spectral sensors, further study needs to be done by integrating hyper-spectral data, which can capture several other water quality parameters that are optically inactive, alongside other deep and machine learning algorithms that were not included in the current study.

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CRediT authorship contribution statement

Vicky Anand: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Bakimchandra Oinam:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Silke Wieprecht:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare no competing interests

Data availability

Sentinel datasets used in this study are available in the public domain through the official portal (<https://dataspace.copernicus.eu/>). ResourceSat datasets used in this study are available in the public domain through the official portal (<https://bhoonidhi.nrsc.gov.in/bhoonidhi/home.html>). The field water quality data used in this

study is under proprietary control can be shared on request. The sample of the field data along with the codes used in this study is accessible in Zenodo repository (<https://doi.org/10.5281/zenodo.13972855>).

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